Project Proposal: Replicating *Detecting Pretraining Data from Large Language Models* (Min-K% Prob)

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Abstract

This proposal sets out a novel replication project based on *Detecting Pretraining Data from Large Language Models* (Shi et al., 2024), published at ICLR 2024. The paper asks whether an outsider with only query access to a language model can tell that a particular text was in its training corpus. It proposes a detection statistic, Min-K% Prob, and a benchmark, WikiMIA, built from Wikipedia articles written on either side of a model's training cut-off. I describe the paper and its evaluation for a general computing reader, justify the choice on venue quality, code and data availability and compute feasibility, and set out the new data the method will be applied to: five existing benchmarks the authors did not use, and two datasets I will build. I also argue the paper's headline claim is worth re-testing rather than merely re-running, since two later papers argue results of this kind are partly an artefact of how the benchmark was built.

1 Introduction

Large language models are trained on corpora that are rarely disclosed in full. That secrecy matters in practice: an author wants to know whether their book was used, a conference whether its benchmark leaked into training, a hospital whether a patient record ended up in a public model. *Membership inference*—deciding whether a given text was part of a model's training set—is the technical question underneath all three.

This project replicates Shi et al. (2024), which brought that question to the pretraining setting and released both a method and a benchmark. It has three phases (Figure 1): reproduce the original results on the original data; apply the released code to existing datasets the authors never used; and build new datasets and test whether the paper's claim survives contact with them.

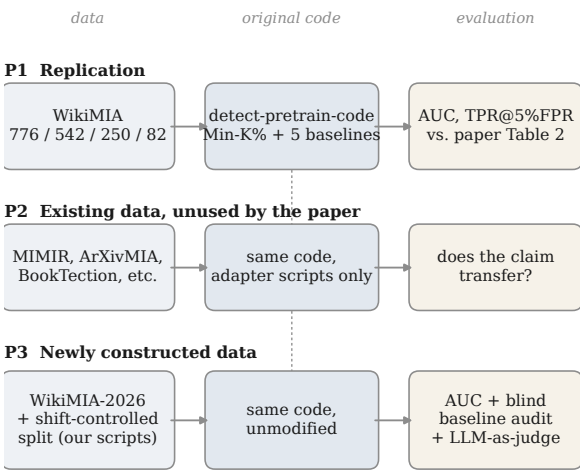


Figure 1: The three phases. The original implementation is held fixed throughout; only the data around it changes. Phases 2 and 3 are the two kinds of new data the assessment requires.

2 Description of the Paper

The problem. Suppose a company releases a language model but not the list of documents it was trained on. Given a sentence x, and enough query access to obtain the probability the model assigns to each word of x, can we decide whether x was one of those documents? This is a *membership inference attack* (MIA), first formalised for classifiers by Shokri et al. (2017).

The intuition is that models predict text they have already seen more easily, so a natural detector is the model's average surprise—its perplexity—with low surprise taken as evidence of membership (Yeom et al., 2018). But surprise also tracks how ordinary the text is. A boilerplate sentence such as "the film received generally positive reviews" is unsurprising to any model, seen or not, while a sentence full of rare proper nouns is surprising even if memorised. Average surprise therefore confuses *memorisation* with *predictability*.

The method. Shi et al. (2024) make a sharper observation: what marks an unseen passage is not that it is uniformly surprising but that it contains *a few* words the model finds very unlikely. Training pushes those outlier probabilities up, so a seen passage tends not to have them. Instead of averaging over all tokens, then, average only over the worst ones. For $x = x_1 \dots x_N$, let Min-K%(x) be the $k\%$ of tokens with lowest model probability; the score is

$$\frac{1}{E} \sum_{x_i \in \text{Min-K\%}(x)} \log p(x_i \mid x_1, \dots, x_{i-1}), \quad (1)$$

where E is the size of that set. A higher (less negative) score means fewer catastrophic surprises and so predicts membership; a threshold turns it into a decision.

As an example, take a ten-token sentence with log-probabilities $[-0.1, -0.2, -8.0, -0.3, -6.0, -0.4, -0.5, -0.2, -1.0, -0.3]$. Its mean is -1.7, which looks poor. But at $k = 20$ the two lowest tokens are -8.0 and -6.0, giving a Min-20% score of -7.0: two badly mispredicted words are dragging the passage down, which the paper reads as evidence the model has not seen it. A more confidently predicted or memorised passage could instead have a much less negative Min-20% score, indicating that even its least probable tokens are relatively unsurprising to the model.

Two properties make this attractive: it is *reference-free*, needing no second model trained on similar data and none of the shadow models Shokri et al. (2017) required, which is what makes it usable when the corpus is secret; and it is cheap, at one forward pass per passage.

The benchmark. The method needs data where membership is known, and the paper's solution is elegant. Take Wikipedia articles about events that happened *after* a model was trained: these cannot be members. Take articles created long before: these almost certainly are. The authors use 1 January 2023 as the cut-off, collecting 394 post-cut-off events as non-members and 394 events from older pages as members (the paper describes these inconsistently as pre-2017 and as pre-2016, which a replication should settle against the released data). Because the split is defined by time rather than by any property of the model, the benchmark—*WikiMIA*—can be reused for any model released in the intervening window, and refreshed as new events occur.

Claims. The paper claims that Min-K% Prob outperforms all previous detectors on WikiMIA with no training; that detection improves with model size and passage length; and that the method is useful for detecting copyrighted books in training data, detecting benchmark contamination, and auditing models that claim to have "unlearned" content.

3 Explanation of the Evaluation

3.1 Metrics

Detection is scored as binary classification over passages, members positive. Since the detector outputs a continuous score rather than a single fixed decision, the paper evaluates it using AUC-ROC and TPR at a low false-positive rate (TPR@5%FPR).

AUC-ROC. The ROC curve plots true positive rate against false positive rate as the threshold sweeps the score range; the area under it summarises the curve in one number (Fawcett, 2006). Its useful reading here is probabilistic: AUC is the probability that a randomly chosen member outscores a randomly chosen non-member, with 0.5 chance and 1.0 perfect. As a worked example, take members m_1, m_2 scoring -2.0 and -3.0 and non-members n_1, n_2 scoring -2.5 and -4.0. Of the four member/non-member pairs the member wins in (m_1, n_1), (m_1, n_2) and (m_2, n_2) and loses in (m_2, n_1), so AUC $= 3/4 = 0.75$. Only the *ranking* matters—rescaling every score leaves AUC unchanged—which is why it is the right summary for attacks whose scores are on incomparable scales.

TPR at low FPR. AUC averages over all operating points, including useless ones. In a privacy setting that misleads: an attack that flags half of all non-members is worthless however high its AUC, because a real accusation of training-data misuse must be made confidently about specific documents. Carlini et al. (2022) argued MIAs should be judged by their true positive rate in the high-confidence regime, and this paper follows that convention with TPR@5%FPR. Operationally, with 394 non-members a 5% FPR permits about 20 false alarms: set the threshold at the 20th-highest non-member score and report the fraction of members above it.

3.2 Datasets used in the paper

WikiMIA is the primary benchmark, released on the HuggingFace Hub under an MIT licence (Shi et al., 2023). It ships as four splits by passage length—32, 64, 128 and 256 words—holding 776, 542, 250 and 82 examples, each a pair `(input, label)` with `label=1` for members. The splits shrink because they are produced by filtering the same article pool for length, so the 256-word split keeps only the longest articles. The code also exposes a paraphrased variant, in which passages have been rewritten by ChatGPT, testing whether the detector tracks the surface string or the content.

BookMIA supports the copyright case study: 9,870 book snippets, with members drawn from pre-2023 books that Chang et al. (2023) had independently shown to be known to GPT-3/GPT-4, and non-members from books first published in 2023. Its ground truth is inherited from a separate paper rather than a temporal rule, which is worth weighing when judging that evidence. The other two studies use data built for the occasion: a *controlled contamination corpus*, on which a model is further pretrained while varying corpus size, target-passage frequency and learning rate; and Harry Potter questions used to audit the "unlearned" model of Eldan and Russinovich (2023).

3.3 Baselines

Five prior detectors are the yardstick a replication must reproduce. LOSS/PPL thresholds perplexity directly (Yeom et al., 2018). ZLIB divides log-perplexity by the text's zlib compression length, using compressibility as a proxy for intrinsic predictability, and LOWERCASE compares the passage's perplexity with that of its lower-cased form (Carlini et al., 2021). SMALLER REF calibrates against a smaller model trained on similar data, so text that is merely easy does not look memorised. NEIGHBOR perturbs the passage into similar variants and asks whether the original sits in a local minimum of the loss surface (Mattern et al., 2023); it is the strongest baseline and much the most expensive.

3.4 Headline result, and why it needs re-testing

Averaged over the WikiMIA settings, Min-K% Prob reaches an AUC of about 0.72 against 0.67 for the best baseline, a relative improvement of 7.4%. That is the number a replication must first reproduce.

It is also worth interrogating, which is much of the paper's appeal. Duan et al. (2024), evaluating models from 160M to 12B parameters trained on the Pile (Gao et al., 2020), found membership inference against LLMs "barely outperform[s] random guessing" once members and non-members are drawn from the same distribution, and argued that apparent success in benchmarks like WikiMIA comes substantially from *temporal distribution shift*: post-2023 text differs from pre-2016 text in vocabulary and style, not only in membership. Das et al. (2024) pressed further, showing that "blind" baselines with no model access—a bag-of-words classifier over the text alone—can match or beat published attacks on several such benchmarks. Meanwhile the method has been extended rather than discarded: Min-K%++ (Zhang et al., 2025) was a Spotlight at ICLR 2025, and calibration-based successors still use WikiMIA as the reference benchmark (Zhang et al., 2024b,a). Replicating this paper therefore lands in a live disagreement, which gives the investigation component of the project real content rather than a re-run.

3.5 Initial thoughts on LLM-as-judge

An LLM-as-judge evaluation puts a strong language model in place of human annotators, scoring outputs by pairwise comparison or a rubric (Zheng et al., 2023). My initial assessment is that it *cannot* replace the primary metric here, but has two genuinely useful auxiliary roles.

It cannot replace the metric because membership is not a property of the text. Whether a passage was in a training corpus is a fact about another model's data pipeline; no amount of reading the passage reveals it. A judge asked "was this in the training data?" would be guessing from surface cues—precisely the failure mode Das et al. (2024) identify—so high judge accuracy would be evidence of a broken benchmark, not a working evaluation. Ground truth here comes from construction (dates, or a documented corpus), and AUC against it remains the metric.

Where a judge does help is, first, in *auditing the data I build*: given a member and a non-member passage with explicit dates removed, ask a judge which is more recent, and how confidently. If it can tell, the split is confounded and any AUC measured on it is partly measuring the confound. That turns the Duan and Das critiques into an automatable check to run over each new dataset before the

attack. Second, a judge suits *quality control on paraphrases*, since that setting depends on rewrites preserving meaning while changing form—a classic judgeable property—and it is reasonable for summarising what distinguishes the passages the attack gets wrong. Two risks need managing: the judge may itself have trained on the post-cut-off articles, contaminating its recency judgements, and LLM judges favour text resembling their own output (Panickssery et al., 2024), which matters where one side is machine-generated. I will therefore report judge agreement against a small human-labelled sample rather than treating the judge as ground truth.

4 Justification for the Choice of Paper

I ranked candidates on whether the code runs, whether the original data is obtainable, whether the compute fits a free Colab/VM budget, and whether the evaluation is reproducible, and only then on venue quality and interest. This paper scores well on all of them.

4.1 Venue and standing

The paper was published as a conference paper at ICLR 2024. ICLR is ranked **A*** by CORE in both the CORE2023 and CORE2021 rounds, under field-of-research code 4611, Machine Learning (Computing Research and Education Association of Australasia, 2023). In the 2025 edition of Google Scholar Metrics it is the **8th**-ranked venue across all disciplines, with an h5-index of 362 and h5-median of 652 (Google Scholar, 2025); the secondary research.com ranking places it 6th in computer science. The authors are based at the University of Washington and Princeton University.

Venue metrics aside, the more informative indicator is that the work has become a reference point in its subfield. WikiMIA is the standard evaluation set in at least four subsequent papers at strong venues—an ICLR 2025 Spotlight (Zhang et al., 2025), an EMNLP 2024 main-conference paper (Zhang et al., 2024b), a COLM 2024 paper (Duan et al., 2024) and a follow-up method paper (Zhang et al., 2024a)—and its central claim has been publicly contested (Duan et al., 2024; Das et al., 2024). A result others have built on *and* argued with is exactly what a replication study can usefully weigh in on.

4.2 Code

The reference implementation is public under Apache-2.0 (Shi, 2023), with 243 stars and 29 forks at the time of writing—evidence of real third-party use rather than an abandoned artefact. It is about 250 lines across `run.py`, `eval.py` and `options.py`, which is a virtue here: I can read all of it and explain every step in the update presentation. Its dependencies are the standard set (`torch`, `transformers`, `datasets`, `scikit-learn`, `matplotlib`, `tqdm`), with no compiled extensions or custom CUDA kernels.

Reading and running the code surfaced four issues, and documenting them is part of the replication. There is no `requirements.txt`, so versions must be pinned by me. `run.py` imports `openai` unconditionally and its OpenAI path targets `text-davinci-003`, an endpoint since retired, so the proprietary-model rows of the paper's tables cannot be reproduced at all—I will say so explicitly rather than quietly omitting them. The scoring convention needs care: `run.py` stores the *negated* mean of the lowest-probability tokens while `eval.py` negates again inside `roc_curve`, so a naive reimplementation would silently invert the ROC curve. Finally, a trap that is in the repository rather than the code: a committed `out/` directory ships results from the authors' own LLaMA-13B run, which any script that globs for output files will happily return in place of the user's own. Catching this class of problem is what separates a replication from a re-run.

4.3 Data and compute

WikiMIA is 469 KB in total, MIT-licensed, and loads in one line via the `datasets` library; it is downloaded on the order of a thousand times a month, so it is live rather than link-rotted. BookMIA is likewise public and MIT-licensed. This satisfies the unit's requirement that datasets not be restricted. The two case studies I cannot reproduce—contamination, needing continued pretraining, and the GPT-3 copyright study, needing a retired API—are ruled out by compute and availability rather than licensing, and I will scope them out explicitly.

My budget is a Macquarie VM plus free-tier Colab (one T4, 16 GB, with session limits). Table 1 maps the paper's target models onto it: I will replicate the small and mid-sized rows in full and state that the 20B–66B rows are out of scope, which the assessment permits provided the limitation is

Model	Params	Status
Pythia-160M/410M	≤0.4B	CPU, minutes
Pythia-1.4B	1.4B	T4 fp16
Pythia-2.8B, OPT-2.7B	2.8B	T4 fp16
Pythia-6.9B, LLaMA-7B	7B	T4 8-bit, tight
GPT-NeoX-20B	20B	out of scope
OPT-66B, LLaMA-65B	65B+	out of scope

Table 1: Target models against a free-tier T4. A full WikiMIA pass is 776 passages × 2 forward passes, so cost is dominated by model loading, not inference.

justified. This is not fatal to the scientific question, since the paper's own scaling claim is that detection *improves* with size, so reproducing the 160M–7B trend tests that claim directly.

The Pythia suite (Biderman et al., 2023) matters for a second reason: it was trained on the Pile (Gao et al., 2020), which is public and documented. For every other model in the paper the membership labels are inferred from release dates; for Pythia they can in principle be *checked* against the actual corpus. That is what makes the shift-controlled dataset of Section 5 possible.

4.4 Preliminary execution check

As required for a novel project, I have downloaded and run the software rather than only reading about it. The repository clones cleanly at commit `a5048f9` and the pipeline runs end to end on the WikiMIA length-32 split *with no changes to the source*—and, as it turned out, under current library versions (`transformers` 5.15.1, `datasets` 4.0.0) rather than the 2023 versions it was written against, which is a better durability result than I expected. On Pythia-160M the full 776-example pass took 1.3 minutes on a Colab T4 and returned an AUC of 0.532 for Min-20% Prob against 0.553 for the perplexity baseline, along with the ROC figure and `auc.txt` the code writes to its output directory.

Two observations follow already. A thirty-line reimplementation of the statistic, written from the paper rather than the code, returns 0.532 on the same data—agreement to three decimals, which confirms the double negation described above has been read correctly. And Min-K% does not beat perplexity here; but at 160M parameters this model is an order of magnitude smaller than the smallest in the paper's own table, and the paper's scaling claim predicts weak detection at this size, so the run establishes that the harness works rather than testing the claim. Scaling up to the models in Table 1 is the first task of the replication proper.

Risks. Gated weights on the Hub are avoided by treating the ungated Pythia and OPT families as primary and LLaMA as optional, and Colab session limits by caching per-model results to disk so runs resume. The remaining risk—that my numbers diverge from the paper's—is not really a risk: given the published disagreement over this benchmark, a well-documented divergence is a result, provided I can show where it comes from.

5 New Data

5.1 Existing datasets not used in the paper

Table 2 lists five public benchmarks that post-date the paper and that Min-K% Prob was not evaluated on in the original work, with an assessment of each. MIMIR is the primary target: built specifically to test the distribution-shift objection, it covers seven Pile domains and is designed around exactly the Pythia models I can run. arXivTection and BookTection are framed around copyright rather than privacy, so they test whether the claim generalises across framings, but both ship in DE-COP's multiple-choice format and need an adapter. PatentMIA is the least feasible and I include it honestly on that basis. The Pile's held-out splits are the most interesting scientifically, because members and non-members there differ by nothing except which shard they landed in.

5.2 Datasets I will construct

ND-1: WikiMIA-2026. The paper describes its benchmark as dynamic, so the most faithful new dataset is to run its construction procedure again for a 2026 cut-off. A Python script will query the MediaWiki API for articles in the relevant year categories, retrieve each page's earliest revision timestamp to establish creation date, strip wiki markup, and truncate into the same 32/64/128/256-word buckets, emitting JSONL in the schema the original code already reads so `run.py` runs unmodified. Because the pipeline is a script, variants are nearly free: I plan a general-events set and a topically restricted one (Australian events) to test whether detection is sensitive to domain as well as date. This is the splicing-Wikipedia recipe discussed in lectures, applied to the paper's own construction.

Dataset	What it is	Access	Feasibility assessment
MIMIR (Duan et al., 2024)	Member/non-member pairs across 7 Pile domains (arXiv, GitHub, Pile-CC, PubMed Central, Wikipedia, HackerNews, DM Mathematics) at three n-gram overlap thresholds	HuggingFace `iamgroot42/mimir`, MIT	**High.** Built for Pythia, so it matches my compute exactly. Stored as paired `member`/`nonmember` columns, so it needs a ∼20-line reshaper into the `(input, label)` schema the code expects. Primary target.
arXivTection (Duarte et al., 2024)	1.55k passages from arXiv articles, labelled by whether the article predates the model	HuggingFace `avduarte333/arXivTection`, MIT	**High.** Small and plain text. Distributed in DE-COP's multiple-choice format, of which only the verbatim passage is used, so an adapter script is required.
BookTection (Duarte et al., 2024)	16.4k rows of passages from 165 books at three passage lengths	HuggingFace `avduarte333/BookTection`, MIT	**Medium.** Same adapter as above. Long passages raise per-example cost, so I would subsample; valuable as a copyright-framed contrast to WikiMIA.
PatentMIA (Zhang et al., 2024b)	Chinese patent abstracts published after the target models' cut-off	GitHub `zhang-wei-chao/DC-PDD`	**Low–medium.** Needs a Chinese-capable target model, changing both the model family and the tokenisation that Min-K% is defined over. Stretch goal only, if earlier phases finish early.
Pile held-out splits (Gao et al., 2020)	The Pile's validation and test shards, contrasted with training shards	HuggingFace mirror `monology/pile-test-val`	**Medium.** Ground truth by construction for Pythia, with no temporal confound at all. The original host has withdrawn the corpus, so I rely on a mirror carrying no dataset card, whose correspondence to the official splits I would need to verify, and on sampling rather than 800 GB. Overlaps MIMIR, which is Pile-derived, but is distinct.

Table 2: Five existing datasets the original paper did not use, with a feasibility assessment of each.

ND-2: A shift-controlled split with a confound audit. ND-1 inherits the weakness Duan and Das identify, so the second dataset removes it. Members and non-members will be drawn from the *same* time window and topical pool, with membership established, where possible, from Pythia's documented training data and training-order information rather than inferred only from publication date; any remaining uncertainty about exact model exposure will be recorded explicitly. Shipped with the dataset is an audit script implementing the blind-baseline test of Das et al. (2024): a TF–IDF and logistic-regression classifier that sees only the text, never the model. If that blind classifier achieves high AUC, the split is confounded and any attack result on it is uninterpretable. Reporting blind AUC beside attack AUC for every dataset in the project—including WikiMIA itself—is, I think, the most useful single contribution this replication can make. The LLM-as-judge check of Section 3.5 gives a second, independent reading of the same question, and a small human-labelled set (about 200 items, annotated by me and two others) will calibrate both.

6 Tools, Plan and Ethics

The project exercises unit techniques across the pipeline: scripted data gathering from the internet (MediaWiki API, HuggingFace Hub); cleaning and curation (markup stripping, deduplication, length normalisation, schema conversion for four external datasets); analysis (ROC/AUC with bootstrap confidence intervals—at 82 to 776 examples per split, a few AUC points may not be meaningful, which the original paper does not report); reproducible environments on a VM and Colab; version control and open science, with a public repository holding the new data, generation scripts, run logs and reports, released in the project GitHub repository; and LaTeX, of which this document is the first instance.

On ethics, the project will use publicly accessible research datasets and openly available sources, while the licence and permitted use of each dataset will be checked and documented before redistribution. Newly constructed data will be drawn from openly accessible sources such as Wikipedia rather than private personal records. The project studies a

Weeks	Milestone
6–7	Pin environment; reproduce WikiMIA for Pythia-160M to 2.8B, all four lengths, all six methods.
8	Add OPT and 8-bit 7B models; bootstrap CIs; compare against the paper's table.
9	Adapters for MIMIR and arXivTection; run phase 2. **Update presentation, 9 Oct.**
10	Build ND-1 and its generation script.
11	Build ND-2 and the blind-baseline audit; run the audit over every dataset.
12	LLM-as-judge checks; human calibration sample; error analysis.
13	Finalise repository and report. **Final report, 6 Nov.**

Table 3: Indicative plan against the assessment deadlines.

privacy attack, but against already-published models and corpora, and for the defensive purpose of auditing whether such attacks work as claimed.

Summary. This is a strong replication target on the criteria that matter: an A* venue, a codebase I have already run end to end, a 469 KB dataset, and two well-understood metrics. It is also one where re-running is not the whole job, since the field has since disagreed about whether the effect is real—which gives the project a question beyond reproduction, and new data of my own with which to answer it.

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